



RISK ASSESSMENT

CLUSTER MENINGITIS C

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SIGNAL

A cluster of meningococcal C ST-11 complex (cc11) has been ongoing since November 2025. To date, 8 confirmed cases have been recorded. The strain is particularly virulent (50% case fatality rate) and, except for a secondary case in the same family, no epidemiological link between the cases has been identified.

DESCRIPTION

Event

On 5 February, the National Reference Centre for meningococcus reported a **cluster of six cases occurring between 11 November 2025 and 25 January 2026**, five of which were identified in January (including one secondary case). Since then, two further cases have been reported (*see epicurve in Annex 1*).

The cases occurred in the 3 regions (1 in Hainaut, Wallonia, 3 in Brussels and 4 in Flemish Brabant, Flanders), but 4 cases are residents or have a connection with 2 **municipalities** (such as attending school or a wedding). The **median age is 45 years** (range 7– 67) (*see distribution by age groups in Annex 2*) and the sex ratio M/F is 0,33. One case involved a homeless person. The strain does not show resistance to antibiotics. The first case was vaccinated against MenC, and investigations did not reveal any underlying immunological deficiency.

Based on the current available evidence **no direct epidemiological link** has been found between the cases, suggesting that the strain is circulating among asymptomatic carriers. However, it was not possible to investigate the settings and groups attended by all cases (particularly in fatal cases).

The cases do not appear to be linked to travel. One case lived in Italy but was supposedly contaminated by a family member in Belgium and died here. Genomic analysis by ECDC on a EU-wide dataset showed two historical cases linked to the same strain: one from France (January 2025) and one from Italy (May 2025).

Type of risk <i>Unexpected & unusual</i>	This event is unexpected and unusual. Meningococcal C has become rare since the introduction of routine vaccination in Belgium in 2002, and the number of cases has further decreased since the COVID-19 pandemic. Between 2023 and 2025, 1–2 cases per year were confirmed in Belgium (all unvaccinated). The age group affected is also unusual (median age is 45). The age groups normally at risk for meningococcal disease are children under 5, young adults aged 19–24 and those aged 65 and above. In meningococcal outbreaks (Men C or B) described in the literature, it is generally young adults who are affected. This age group might be less affected in the current outbreak given the generalised vaccination programme introduced in 2002. The ST-11 cc11 has been associated with various serogroups, including MenC, MenW and, less often MenB, (1–6). This specific clonal complex is characterized and associated with large outbreaks, high case fatality rates (usually 18-21% lethality but it could be over 30%) and atypical presentations (i.e., joint infection, gastrointestinal symptoms, and urethritis). Outbreaks related to this clonal complex affect multiple populations, including educational institutions, men who have sex with men (MSM), Hajj pilgrims and military personnel.
Severity of the risk <i>High</i>	The severity of the risk is assessed as high: invasive meningococcal infections are severe and the particular strain involved in this cluster is highly virulent with a case fatality rate of 50%, which is much higher than the usual rate (5-10% of lethality).
Exposed population	More information is needed about settings, risk factors, etc. to better define the exposed population. Based on the currently available epidemiological information, adults aged between 30 and 70 seem the group most affected by this outbreak. Although the vaccination status of most cases is unknown, they are probably unvaccinated given their year of birth. Routine vaccination against meningococcal C began in 2002, with a catch-up vaccination program for people born between 1996 and 2001 in Wallonia and Brussels, and between 1982 and 2001 in Flanders. Coverage rate for this catch-up vaccination program is unknown but in Flanders, vaccination was carried out via the school medicine and achieved a high uptake. Those who have been vaccinated as infants appear to be currently protected. According to the latest vaccination coverage survey, coverage against meningococcal C (for children aged 15 months) is more than 90% in the 3 regions (Flanders : 92,9% in 2020, Wallonia : 90,6% in 2019 and Brussels : 92,2% in 2019). However, the first case in the cluster is a case of vaccine failure in a 7-year-old child. In a previous MenC outbreak in Tuscany involving a highly virulent strain, several vaccine failures were reported (7). Twelve cases of vaccine failure were reported in various age groups (out of 61 cases), with vaccine effectiveness in this outbreak estimated at 77%. In the same publication, vaccine effectiveness is estimated at 100% outside the outbreak period (between 2005 and 2016). Protection against serogroup C invasive meningococcal disease (IMD) is primarily mediated by serum bactericidal antibodies (SBA), with defined correlates of protection (e.g., rSBA $\geq 1:8$ or hSBA $\geq 1:4$). Sustained circulating antibody levels are important because disease progression is rapid and may outpace memory responses alone. In a systematic review published in 2024, Griskaitis et al. confirm high protective immunity after a first dose of MenC conjugate vaccines, but subsequent waning over a period of 6 years, with only 20% of participants in booster trials still showing protective antibody titers at pre-booster timepoint (8). For MenACWY vaccines, the Summary of Product Characteristics (9,10) describes

Risk of (inter)national dissemination

Low

long-term immunogenicity data following primary vaccination, with follow-up available up to 10 years for Nimenrix (TT) and up to 5 years for Menveo (CRM), however, the duration of protection against Men C has not been fully established. Nevertheless, available surveillance and observational data support good effectiveness of MenACWY vaccines in maintaining control of meningococcal serogroup C disease (8,11).

Since 1 May 2026, no more monovalent meningococcal C vaccine is currently commercialized in Belgium.

It is possible that the number of cases is underestimated, as some strains or clinical samples are not sent to the National Reference Centre. Whole-genome sequencing (WGS) is required to determine whether a strain belongs to the cluster. The NRC coverage (= the number of samples received/total number of reported cases) in 2025 is estimated at 82%: out of a total of 87 cases identified through both mandatory notification and the NRC, 10 samples from Wallonia, 3 from Brussels, and 2 from Flanders were not submitted to the NRC.

The risk of (international) dissemination is considered **low**: meningococcus is moderately contagious and transmission typically requires face-to-face contact within one metre for at least an hour.

According the PubMLST database¹, using a 7 AD threshold, the Belgian MenC isolates cluster with isolates from:

- Canada: N=24 (2022–2026, unknown clinical presentation).
- The Netherlands: N=1 (2023, unknown clinical presentation)
- UK (Scotland): N=1 (2023, invasive).
- France: N=1 (2025, invasive)
- Italy: N=1 (2025, invasive)

At present, the cluster thus appears to be largely confined to Belgium.

Meningococcal infections occur most frequently between December and March. There may therefore be a slight reduction or halt in the number of cases over the coming months, but a rise is likely to be seen again next autumn and winter.

PREPAREDNESS & CONTROL MEASURES ALREADY IN PLACE

Preparedness

- **Surveillance:**
 - Invasive meningococcal infections are **notifiable** to the regional authorities.
- **Management of cases and close contacts:**
 - For each reported case, **antibiotic prophylaxis** is administered to close contacts. Furthermore, **vaccination** of close contacts is recommended in the event of an isolated case (Men C, W or Y only) or a cluster (Men B, C, W, Y). This latter recommendation had already been in place in Flanders for several years and was introduced in Wallonia and Brussels in March 2026.
- **Diagnostic:**
 - The NRC, hosted by Sciensano, receives and analyses all samples from suspected or confirmed cases of invasive meningococcal disease that are submitted on a voluntary basis. All cultivable isolates undergo whole genome sequencing (WGS), and the resulting data are used for detailed strain characterization and assessment of genetic relatedness. This work is carried out on a weekly basis, enabling real-time detection of clusters and outbreaks. In addition, WGS data are

¹ PubMLST is a public, database hosting molecular typing and whole-genome sequence data for multiple bacterial species, including meningococci, allowing strain comparison and nomenclature.

	shared monthly with the ECDC through the international EMERTII database managed by the agency, allowing for the detection of clusters beyond national borders. • Prevention: ○ In Belgium, a vaccination program for children aged 15 months has been implemented by regional authorities, initially against MenC and since 2023 against MenACWY. The protection declines however over time. Adolescents and young adults are the main carriers and primary transmitters of meningococcal bacteria. Since 2019, the Superior Health Council has therefore recommended routine MenACWY vaccination for adolescents. This vaccine is currently not included in the vaccination program of the federated entities, meaning it is not provided free of charge and there is no structured schedule for its administration. MenACWY vaccines are available for purchase at the pharmacy. Currently Belgium has a stock of 23,000 vaccine doses, which covers around 8 months of usual demand. Such a quantity is very limited for any targeted vaccination campaign.
Specific Control Measures	• Communication about the cluster and, most importantly, a reminder to send all clinical samples to the NRC was sent out via the Flash Infectious Diseases in early March. For each new confirmed or probable case, information is shared between the regions, the NRC and Sciensano's epidemiologists. • An alert was issued to other European countries in February via the EpiPulse communication platform to determine whether the same strain identified in our cluster is circulating elsewhere in Europe.

PUBLIC HEALTH IMPACT IN BELGIUM	
Low	• Public health impact is considered as low because of the low number of cases. However, severity of the strain is very high with a high probability of death. • Meningococcal infections are a major public health concern. The issue is particularly sensitive in the wake of the outbreak in the United Kingdom, and it has a substantial psychological impact on the general public.

RECOMMENDATIONS	
	Communication • To GPs to inform them about this outbreak and to reiterate the SHC's recommendations on vaccinating people at risk against Men ACWY (e.g. splenectomised individuals, immunocompromised persons, etc.). • To laboratories to remind them of the importance of sending every strain of invasive meningococcus to the NRC for confirmation. Laboratories should not wait for culture but can already send a sample after a positive PCR. • To hospitals to raise awareness, emphasize the importance of swift diagnosis and notification and repeat recommendations on post-exposure prophylaxis for close contacts. • To the general population: at this stage, a proactive communication to the general public is not recommended as there are no actionable recommendations for them. However, a harmonized communication

should be prepared already, to be used by all stakeholders in case the situation would change.

Diagnostic:

- It is essential that **all meningococcal strains/clinical samples are send to the NRC**, and that regional authorities ensure this is systematically followed up for each reported case.

Epidemiological investigation:

- Develop a **case definition** and **standardized questionnaire** for cases.
- Maximize available information for cases (using the questionnaire and, where possible, by recontacting past cases or their families).

Vaccination:

- **Mass (or targeted) vaccination** is currently not recommended, with the following rationale:
 - If contact tracing and chemoprophylaxis are insufficient, mass vaccination has been successfully used in several meningococcal outbreaks (5,12–16). Both MenC and MenACWY vaccine are highly effective in reducing disease in vaccinated groups and in producing herd immunity in unvaccinated cohorts.
 - Some advisory bodies, like ACIP in the US, recommend a threshold of **>10 cases per 100,000 population over a 3-month period** (17) to trigger mass vaccination. In the current outbreak, this threshold is as of yet not met.
 - Additionally, there is a lack of epidemiological information to clearly define the population that should be vaccinated.
 - However, it is recommended to start considering implementation aspects of a potential future mass vaccination campaign in order to be prepared in case of an increase of cases.
- In general, implementation of **universal vaccination of adolescents** against MenACWY (as recommended by the Superior Health Council) could help prevent such outbreaks by boosting immunity and, in particular, reducing carriage and transmission among adolescents and young adults.

Mass chemoprophylaxis (i.e., administration of antibiotics to substantial populations) is not recommended to control large outbreaks of disease. Multiple sources and prolonged risk for exposure make this approach impractical and unlikely to succeed (17).

ACTIONS

- Prepare communication for GPs: Sciensano
- Send communication to GPs: federated entities
- Prepare communication for laboratories: Sciensano
- Send communication to laboratories: Sciensano
- Prepare communication for hospitals: Sciensano
- Send communication to hospitals: RMG
- Implement SHC recommendation for adolescent vaccination: federated entities
- Epidemiological investigation :
 - Develop case definition/standardized questionnaire: Sciensano
 - Collect maximal available info from cases : regional health authorities
- Reflect on potential targeted vaccination campaigns: federated entities/RMG

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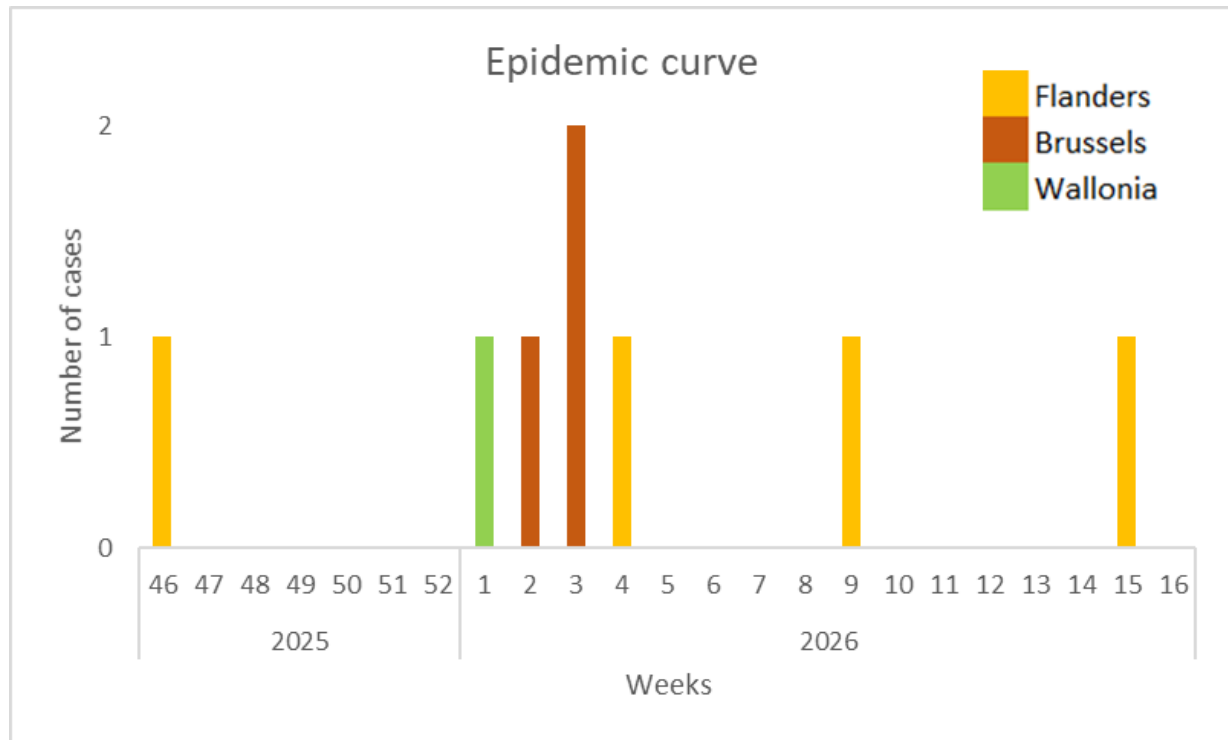
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ANNEXES

ANNEX 1 : EPIDEMIC CURVE



ANNEX 2: CASE DISTRIBUTION BY AGE GROUP

Age group (year)	Number of cases
<10	1
10-19	0
20-29	0
30-39	2
40-49	2
50-59	1
60+	2

ANNEX 3: METHODOLOGY

DESCRIPTION METHODOLOGY	
EVIDENCE	
AVAILABILITY OF EVIDENCE	A short literature search was carried-out about control measures, particularly vaccination strategies, implemented for recent serogroup C and B meningococcal outbreaks in European countries and the US. Available epidemiological evidence about the Belgian cases was limited.
EXPERTS	
SELECTION PROCESS	Experts for this risk assessment were selected based on expertise, current position, or as representatives of permanent RAG members. The full list of RAG experts is available at https://www.sciensano.be/en/rag-experts
EXPERT LIST	Caroline Boulouffe (AVIQ), Steven Callens (U Gent), Paloma Carrillo (ONE), Laura Cornelissen (Sciensano), Inne Crevecoeur (FAGG/AFMPS), Iris De Schutter (DZ), Wouter Dhaeze (DZ), Stéphanie Jacquinet (Sciensano), Jamila Hamdani (FAGG/AFMPS), Naïma Hammami (DZ), Wesley Mattheus (Sciensano), Ilse Peeters (Sciensano), Florence Rolin (Vivalis), Petra Schelstraete (UZ Gent), Anna Schmelz (DGOV), Jorgen Stassijns (Sciensano/RAG), Stefan Teughels (Domus Medica), Heidi Theeten (DZ), David Tuerlinx (CHU – UCL Liège), Pierre Van Damme (U Antwerp), Margaux Vermeulen (Sciensano)
DECLARATION CONFLICT OF INTEREST (DOI)	All experts have submitted a DOI. No potential conflict of interest was wound. For those who have not yet submitted, a DOI will be asked.
CONSULTATION PROCESS	
MEETING	Draft Risk Assessment sent to experts on 20/4/2026. Expert online meeting on 22/4/2026 Final draft approved via mail on 4/5/2026 Final Risk Assessment published 6/5/2026
SURVEY	No survey was carried-out.
DECISION PROCESS	
CONSENSUS	Consensus on the final version was reached on 4/5/2026
OBJECTIONS	No major objections were expressed by the experts